

Beyond the Score: A Systematic Literature Review of Explainable Artificial Intelligence Frameworks for Promoting Equity in Automated Educational Assessment

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Abstract

The application of Artificial Intelligence (AI) in educational assessment has revolutionized the way educational institutions grade and provide feedback. Automated assessment systems promise significant gains in efficiency and scalability, but concerns have been raised about transparency, fairness, and algorithmic bias. In particular, concerns have been raised about the extent to which learners and educators can understand, justify and contest automated decisions when opaque ‘black box’ models are used. This work is a systematic literature review of Explainable Artificial Intelligence (XAI) frameworks and their potential to support the development of equity in automated educational assessment systems. We systematically searched peer reviewed studies from 2020 to 2025, using pre-defined inclusion and exclusion criteria. Through a rigorous screening process 25 studies were selected for detailed analysis. Their findings suggest that historical training data can encode algorithmic bias within automated assessment models, which can marginalize learners from diverse linguistic, cultural and socioeconomic backgrounds. Moreover, the application of XAI techniques, such as visualization of feature importance, decision trees and explanation interfaces, was found to further enhance the transparency and trust of the stakeholders. We however find that explainability is not sufficient for fairness. The literature places a great deal of emphasis on Human-in-the-Loop (HITL) frameworks, where decision-making authority remains with educators, and AI systems function as decision support tools. The authors conclude that successful implementation of the AI-assisted assessment depends on a combination of technical transparency, ethical governance and human oversight. Educational institutions, especially in the Philippine context, are recommended to promote equitable and accountable AI use in assessment practices.

Keywords: Explainable Artificial Intelligence, Automated Assessment, Educational Equity, Algorithmic Bias, Human-in-the-Loop, Educational Technology

1. Introduction

AI technologies are increasingly being employed in educational settings, particularly in the areas of assessment, feedback generation and learning analytics. Today educational institutions use automated

assessment systems for assessment of written work, formative feedback and support decision making. The need for scalable education solutions, rising class sizes and administrative burdens have accelerated the use of these technologies.

AI-assisted assessment has been linked with significant benefits but also raises concerns about fairness, accountability and transparency. Many of today's assessment systems are built on complex machine learning and deep learning models, whose internal decision-making processes are not readily interpretable by educators or students. These systems are often referred to as "black-box" models because the way in which outputs are produced is essentially opaque to the end-users.

Black box algorithms are not transparent and this poses serious pedagogical and ethical challenges. When an automated system generates a score or provides evaluative feedback, stakeholders may not know the specific reasons for that determination. This may restrict the possibilities for meaningful feedback, pedagogical reflection and appeal. The lack of interpretability may reduce trust in automated assessment systems and limit their potential to support student learning.

Concerns about algorithmic bias have also received growing scholarly attention. Machine learning models are often trained on historical data sets which may contain pre-existing social, cultural, linguistic or demographic biases. Automated assessment systems could therefore perpetuate or even worsen such biases. This implies that assessment models trained on small or non-representative data can have a disproportionate impact on linguistically and culturally underrepresented students.

Explainable Artificial Intelligence (XAI) has become a promising research area to address such problems. XAI aims to make AI systems more transparent and interpretable by providing understandable explanations for algorithm decisions. Techniques such as feature attribution, visual explanations and interpretable model architectures can enable stakeholders to gain a better understanding of how the assessment outcomes are produced. Greater transparency has been associated with increased trust, accountability and fairness in AI-enhanced educational settings.

As interest in XAI increases, many questions still remain about the effectiveness of XAI in the promotion of educational equity. Explainability can improve transparency, but does not necessarily eliminate bias and fairness. Finally, the extent to which XAI facilitates meaningful human oversight in educational assessment remains an open area of inquiry.

From the existing literature, it is noted that algorithmic fairness and educational technology have received considerable attention. However, there is a lack of synthesis on the specific role of Explainable Artificial Intelligence to promote equitable automated assessment practices, especially in multilingual and culturally diverse educational settings. The gap is especially relevant in developing countries such as the Philippines, for which empirical evidence is still limited.

Thus, the goal of this systematic literature review is to explore the current body of research on Explainable Artificial Intelligence frameworks and their contribution to fostering equity in automated educational assessment. The study's specific objectives are to:

- Think about examples of algorithmic bias in automated assessment systems for education.
- Assess the effectiveness of Explainable Artificial Intelligence (XAI) methods to enhance transparency, accountability, and trust among stakeholders; and
- Identify evidence-based frameworks for equitable and human-centered AI implementation in assessment in education.

The results of this review are expected to add to the continuing discussion on the ethical use of AI in education, and to offer practical recommendations to educational institutions that want to implement trans-

parent, accountable, and fair assessment systems.

2. Related Literature and Theory Background

2.1 Artificial Intelligence for Assessment in Education

Artificial Intelligence (AI) has been achieving a substantial presence in educational settings, especially in assessment, generation of feedback, learning analytics, and prediction of student performance. AI-assisted assessment systems have been developed to make grading more efficient, to reduce the workload of educators and to give timely feedback to learners.

Automated assessment technologies range from rule-based systems and automated essay scoring tools to advanced machine learning and deep learning models that can assess complex student outputs. This has enabled educational bodies to handle large assessments and to maintain consistency in the scoring procedures.

However, automated assessment systems have been criticized for lack of transparency and fairness. It has been noted that very complex AI models act as ‘black boxes’, meaning that it is difficult for educators and students to understand how assessment decisions are made. Questions of responsibility and educational validity have therefore come under attack.

2.2 Algorithmic Bias and Fairness in Educational AI

One of the biggest problems with implementing AI in education has been algorithmic bias. Bias may be introduced at the level of data collection, model development, or system implementation. Machine learning algorithms learn patterns from historical data. Automated decision-making processes may reinforce existing social and institutional inequalities.

It has been suggested that educational data sets contain unconscious bias about language, culture, socio-economic status, gender and ethnicity. Training assessment models on these datasets may lead to inequitable outcomes for some populations of students.

The issue of algorithmic fairness has now become a central theme in the current discourse on educational AI. Fairness is commonly defined as the fair treatment and impact of automated systems on different groups of learners. However, fairness is still a difficult target to achieve, because educational contexts are complex per se and they are affected by a number of social and cultural factors.

Some authors have argued that fairness cannot be measured only by statistical performance metrics. What we need to do is consider wider educational issues such as inclusivity, transparency and the learner experience when assessing the effectiveness of AI systems.

2.3 Explainable AI (XAI)

The disadvantages of black box AI systems have led to the design of Explainable Artificial Intelligence (XAI). XAI stands for Explainable AI. It is a set of methods and techniques that help to make the results of decisions made by algorithms understandable and interpretable for humans.

Traditional machine learning models tend to optimize predictive accuracy, whereas XAI aims at finding a trade-off between model performance and interpretability. Use explainability mechanisms to understand what factors drive automated decision-making.

The popular XAI methods are:

- Feature importance analysis
- Decision tree visualization
- Attention mapping
- Rule extraction techniques

- Model-agnostic explanation methods
 - Natural language explanations

In educational testing these may provide educators and students with a better understanding of how scores and feedback are generated. Transparency has been linked with increased trust, accountability and acceptance of AI-assisted systems.

But explainability is not equalism. Explanation can show how a decision was made, but not whether it was a fair decision. Therefore, we argue that explainability and fairness should be seen as complementary, not alternative, goals.

2.4 FATE Framework

We propose the FATE framework (Fairness, Accountability, Transparency, and Ethics) for responsible AI governance. The framework lays down guiding principles for the development and implementation of AI technologies across different sectors including education.

Fairness is the just treatment of people and groups in automated systems. Accountability refers to the responsibility of organisations and developers for the impact of AI technologies. Transparency means being able to explain how the system works and why it makes the decisions it makes, in a way that can be understood. Ethics deals with the broader questions of human rights, social responsibility and the public interest.

The FATE framework is finding growing application in evaluating the appropriateness of AI-assisted grading systems in educational assessment. We call on educational institutions to make sure assessment technologies adhere to these principles for responsible innovation.

2.5 Human-in-the-Loop Evaluation Models

The Human-in-the-Loop (HITL) frameworks have been one of the most popular ways to cope with the ethical concerns around automated assessment. AI is used in this situation as a decision support tool, and the responsibility for the assessment results remains with the human teacher.

The HITL approach is a hybrid of the pros and cons of human judgment and machine intelligence. Automated systems can process large quantities of data and identify patterns, but educators provide contextual understanding, professional expertise, and ethical judgment.

Research in the area of AI-aided assessment has repeatedly shown that significant human oversight increases trust from stakeholders. Involving educators also helps to identify and fix algorithmic mistakes that would otherwise go undetected.

Therefore, HITL frameworks are increasingly seen as essential components for responsible deployment of AI in education.

2.6 Students' Belief and Acceptance of AI-Supported Assessment

Students' perceptions of educational technologies determine their success. Learners' trust in assessment systems has an impact on the extent to which they engage with, accept and act on feedback.

Students are more likely to trust AI-supported assessment if the assessment procedures are transparent and understandable, research has shown. Opaque systems, on the other hand, can foster perceptions of doubt, anxiety and unfairness.

Explainable feedback has been associated with learner confidence and understanding of assessment results. When students are explaining they can see where they can improve and this can lead to more meaningful learning experiences.

But there are concerns about the over-reliance on automated evaluation systems. A few academics have warned that students might simply adapt their work to the algorithm's preferences, rather than

demonstrating genuine creativity and critical thinking. So, the big fear is the tension between efficiency and educational integrity.

2.7 Theoretical Framework

The review is based on the assumption that a fair AI-assisted assessment needs the interaction of three interdependent components:

- Transparency via Explainable Artificial Intelligence (XAI);
- Accountability, ethical governance and principles of fairness; and (3)
- Human in the loop professional oversight frameworks.

The conceptual basis of the study is the correlation between these components. Transparent AI systems, along with ethical governance structures that allow for meaningful human participation in assessment decision making processes, are argued to improve educational equity.

Figure 1. Conceptual Framework of Equitable AI-Assisted Assessment



3. Research methodology

3.1 Study Design

We adopted the Systematic Literature Review (SLR) approach to synthesize contemporary evidence on the use of Explainable Artificial Intelligence (XAI) in automated educational assessment. The SLR methodology was chosen as it allows for the structured identification, assessment and synthesis of existing research with lower selection bias. Such a method was used in several studies to analyze trends, challenges, and new frameworks on algorithmic transparency and educational equity.

The review was conducted according to principles generally used in systematic evidence synthesis and was guided by the Preferred reporting items for systematic reviews and meta-analyses. Relevant and high-quality studies were included through a systematic screening and selection process.

3.2 Sources of data and search strategy

A literature search was performed using major academic databases and repositories of scholarly work. These were:

- Springer Link
- ScienceDirect Elsevier
- IEEE Xplore
- Google scholar
- Scitpress
- ACM Digital Library Publication

The search was limited to the period 2020 – 2025. The reason this period was chosen is that it was a time of significant advances in the fields of educational AI, explainability methods and generative AI technologies. Previous studies were considered less representative of the current technology environment with deep learning architectures and large language models.

Boolean search operators were used in conjunction with technical and educational terms. The search strings used included:

- ("Explainable Artificial Intelligence" OR "XAI") AND ("Educational Assessment")
- ("Algorithmic Bias" OR "Fairness") AND ("Automated Grading")
- "Human in the loop" AND "Educational technology"
- ("AI Assessment") AND ("Transparency" OR "Accountability")
- ("Educational Equity") AND ("Explainable AI")

Additional studies were identified via backward citation tracking and reference list checking.

3.3 Criteria for Inclusion and Exclusion

Inclusion and exclusion criteria for relevance and quality were predefined.

Eligibility Requirements

Studies were eligible if they:

1. Published between 2020 and 2025;
2. Published in peer-reviewed journals, conference proceedings or in scholarly publications;
3. Explored Explainable Artificial Intelligence, algorithmic fairness, transparency or bias in educational assessment;
4. Studied automated grading, automated feedback systems, or AI-assisted educational decision-making;
5. Discussed implications for students, educators or educational institutions.

Exclusion Criteria

Studies were excluded when:

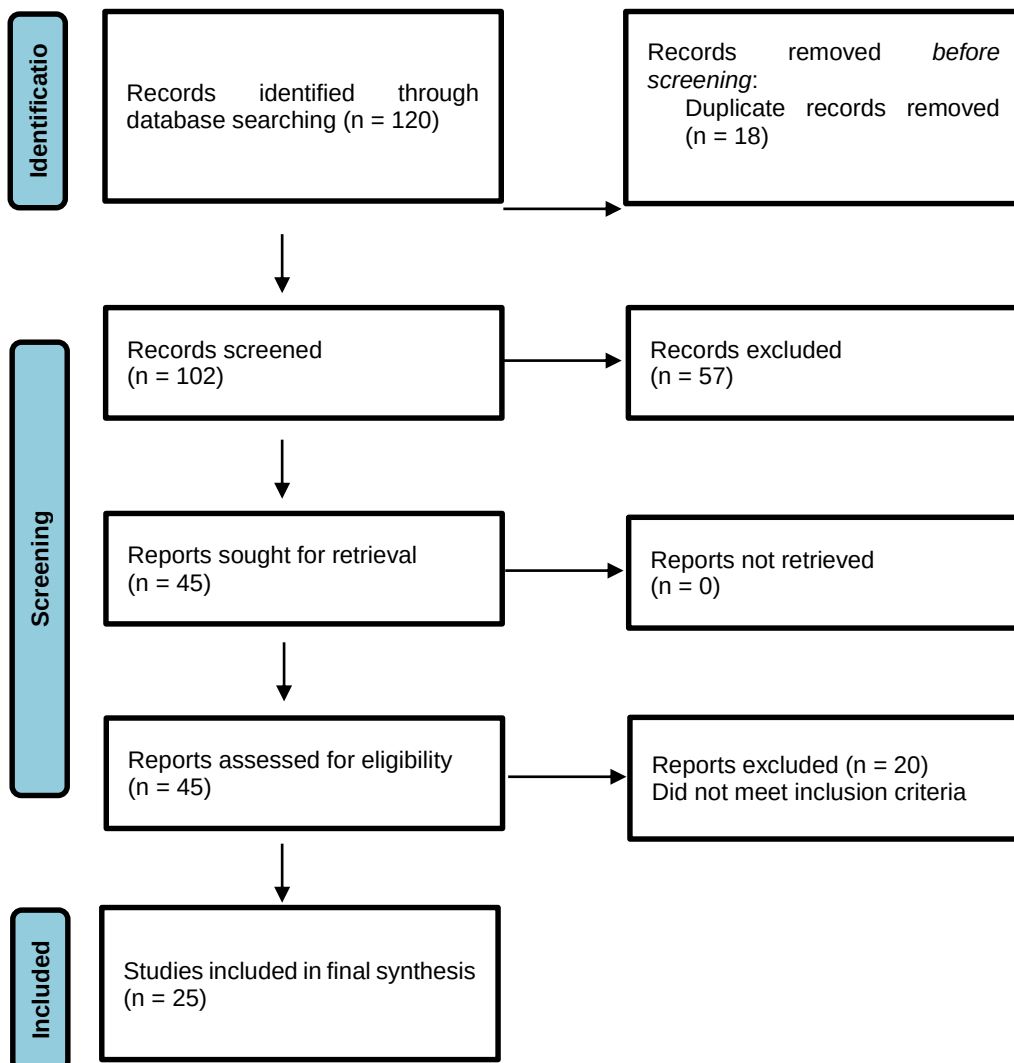
1. Concentrated solely on technical model development, without educational implications;
2. Reviewed other AI applications not related to assessment or feedback;
3. Editorials, opinion papers, blog posts or non-peer reviewed sources;
4. Were duplicates retrieved from multiple databases;
5. Insufficient methodological or empirical detail

3.4 Selection of studies

The first search identified about 120 potentially relevant publications. Before the screening, duplicates were removed. Titles and abstracts were reviewed for relevance to the research objectives.

All studies meeting the initial criteria were reviewed in full text. Inclusion and exclusion criteria led to 25 studies chosen for further analysis.

Fig. 1. PRISMA Flow Diagram for Selection of Studies



The systematic identification, screening, eligibility assessment and selection of studies included in this review are demonstrated in Figure 1. A total of 120 records were retrieved from database searching. Twenty-five studies were eligible and included in the final synthesis following duplicate removal, screening and eligibility assessment.

3.5 Data Extraction and Data Analyses

To ensure consistency in the study, a structured data extraction process was used. The articles offered the following information:

- Author(s) and publication date
- Research Goals
- Educational context
- XAI or AI methodology
- Reported benefits and drawbacks
- Fairness and equality issues
- Recommendations for educational praxis

Extracted data were analysed by thematic synthesis. Recurring themes were grouped for ease of comparison across studies.

The analysis revealed five main themes:

- Algorithmic Bias in Computer-Based Assessment
- Transparency and Explainability Tools
- Human-in-the-Loop Frameworks
- Psychological effects and trust in students
- Ethics in Governance and Accountability of Institutions

Thematic analysis was used to identify common patterns and points of consensus across the literature, as well as highlighting unresolved challenges and future research directions.

3.6 Review Limitations

Some caveats deserve mention. First, the review was limited to studies published in English, and relevant studies published in other languages may have been missed. Second, the majority of the literature is from North America, Europe and China. There is scant empirical evidence from developing countries and multilingual educational contexts.

No studies were found in Philippine educational settings. Thus, conclusions on local implementation were largely based on the more general international evidence and theoretical postulates rather than on the more extensive empirical evidence specific to the Philippines.

Finally, the field of Explainable Artificial Intelligence is developing and new studies may be published after the completion of the review. The selected literature, however, provides a broad overview of the current state of research on explainability, fairness, and educational assessment.

4. Results and Discussion

The review of the selected studies suggested that the use of Artificial Intelligence in educational assessment has both promising opportunities and important ethical concerns. AI-assisted assessment systems have shown the promise of improving efficiency, consistency and scalability but concerns regarding transparency, fairness, accountability and inclusivity are still being reported in the literature.

The analysis identified five major themes: (1) Algorithmic bias in automated assessment, (2) Explainability and transparency mechanisms, (3) Human-in-the-Loop frameworks, (4) Student trust and psychological effects, and (5) Ethical governance and institutional responsibility.

4.1 Algorithmic Bias in Computerized Assessment

One of the most common concerns in the literature reviewed is the existence of algorithmic bias in automated assessment systems. As is well known, the performance of machine learning models is highly

dependent on the quality and representativeness of training data. Automated systems can thus continue bias in historical data sets.

Automated assessment models have been shown to inadvertently favour certain linguistic patterns, writing styles or communication structures that are more frequent in training data. This suggests that students whose language use does not conform to dominant academic language norms may be at a disadvantage in automated evaluation processes.

Several studies have shown the results of automated scoring are subject to socio economic status, ethnicity, cultural background and variation in language biases. In the education datasets where there is a lack of diversity, the assessment systems could misinterpret the non-standard but valid expressions as lower quality in terms of academics.

The results also show that bias is often hard to detect when assessment decisions are made using very complex deep learning architectures. In such cases, the reasoning for a particular score may not be readily accessible to teachers or students. This means that potentially unfair decisions may go undetected.

The literature repeatedly notes that a high predictive accuracy of an assessment system is not enough to declare the system algorithmically fair. A model can be statistically robust but still produce inequitable results for certain groups of learners. Fairness must therefore be evaluated independently of accuracy.

Table 1: Trends in Ethics Issues in Automated Educational Assessment

Era	Dominant Technology	Primary Evaluation Metric	Major Ethical Concern
2010–2018	Automated Essay Scoring	Keyword Density and Grammar Accuracy	Manipulation through keyword optimization
2019–2022	Deep Learning Models	Predictive Accuracy	Hidden algorithmic bias
2023–Present	Large Language Models	Contextual Understanding and Fluency	Hallucination, opacity, and cultural bias

With the advancement of technology, ethical concerns have evolved. These early systems could be “tricked” by a shallow pattern in the language. The challenges today are more about explainability, fairness and reliability.

4.2 Explanation and Transparency Obligations

Explainable artificial intelligence has been suggested to deal with the problems of black box assessment systems. Explainability was one of the most frequently cited key requirements for the responsible use of AI in education across all studies analyzed.

XAI techniques are designed to produce interpretable explanations of the generation of assessment results. Such explanation types can be visualized using feature importance plots, attention maps, decision trees, confidence indicators or textual justification approaches.

Literature suggests that explainability can serve several educational purposes. First, transparency allows educators to evaluate the rationality and pedagogic appropriateness of the automated recommendations. Secondly, students have a greater awareness of the factors that contributed to the results of assessments. Third, explainability enables accountability – it lets us identify and correct potential errors or biases.

But the results suggest that explainability alone is not sufficient to ensure fairness, even with these benefits. An explanation might show how a decision was made, but not that the decision made was educationally

sound. Transparency is thus to be seen as a necessary but not sufficient condition for equitable assessment. Further work also highlighted the importance of explanations being understandable for non-technical stakeholders. Explanations may be complex mathematical representations useful to AI developers but of limited value to educators and students. Explainability must be technically transparent and pedagogically interpretable for it to be useful.

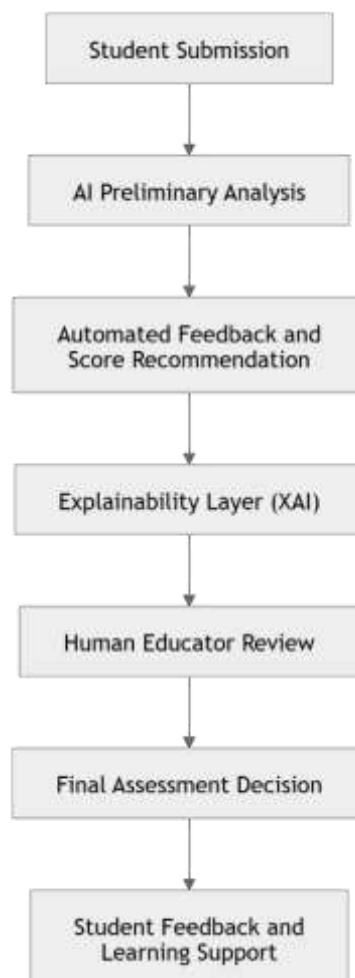
4.3 Frameworks with Human-in-the-Loop

The most supported strategy to promote fairness in AI-assisted assessment among the themes identified in the literature was the Human-in-the-Loop (HITL) framework.

The HITL approach keeps AI systems as decision support tools rather than autonomous evaluators. Automated initial scoring, pattern detection, feedback generation or flagging of potential issues are possible. But the ultimate decision still rests with humans.

The literature indicates that there are several advantages to human oversight. Educators can also take into account contextual factors that may not be visible to algorithms. Interpretation of creative expression, cultural variation and learner-specific circumstances is more appropriate for human judgment. Teachers, of course, always have the option to question or override automated recommendations when appropriate. In the literature we reviewed, fully automated assessment systems were more likely to cause concern among stakeholders than those with meaningful human oversight. Continuous engagement in decision making processes appears to boost confidence in assessment aided by AI.

Figure 2. Human-in-the-Loop Evaluation Workflow



This framework places AI as an assistive technology, not a replacement for professional educational judgment.

4.4 Psychological Effects and Students' Trust

The psychological implications of the automated assessment are also discussed at length in the papers reviewed. Results showed students' perceptions of fairness, autonomy and trust had a significant impact on acceptance of AI-assisted assessment technologies.

Students were seen to talk about their worries about not understanding the basis for automated decisions. Receive uncertainty about grading procedures without explanation. This uncertainty can lead to a decrease in engagement and a loss of confidence in the assessment results.

But research shows open channels of feedback can build students' trust. Students feel more comfortable with automated systems when they are given understandable explanations for assessment decisions.

The literature also suggests that explainability can increase learner agency. Students are not left with arbitrary judgments of the assessment results but they can understand where they need to improve. This process helps in creating a more meaningful learning experience and helps in achieving the goals of formative assessment.

What is worrying is that students write to satisfy the "algorithmic palate" and not for communication. This can thus lead to a conformity at the expense of creativity and critical thinking by over-reliance on automated evaluation systems.

4.5 Leadership Ethics and Institutional Responsibility

The results point out that responsible adoption of AI is not only about the design of technology, but also about the institutional governance structures. Educational institutions will increasingly likely develop policies on the development, procurement, implementation and monitoring of AI systems.

Some studies have pointed to the need for transparent policies that require disclosure when AI tools are used in assessment processes. These policies enhance procedural fairness by informing students about how decisions are made about their assessment.

The literature also recommends multidisciplinary governance committees for the review of applications of AI in education. Such committees could include educators, administrators, technology experts, ethicists and student representatives.

Special attention should be given to the evaluation of AI systems for multilingual and culturally diverse educational settings. Institutions in the Philippines need to validate assessment technologies with local linguistic practices before implementation.

Table 2: Comparison of bias mitigation methods

Strategy	Mechanism	Effectiveness	Limitation
Data Balancing	Diversification of training datasets	Moderate	Difficult to represent all populations
Post-hoc XAI	Generation of explanations after decisions	High	May not fully reveal underlying bias
Human-in-the-Loop	Human validation of AI recommendations	Highest	Requires additional time and resources
Adversarial Learning	Removal of sensitive attribute influence	Moderate to High	Technically complex implementation

Taken together, the evidence from the synthesized studies suggests that Human-in-the-Loop approaches are, at this point, the most reliable method to balance efficiency, transparency and fairness in educational assessment systems.

5. Conclusion and Recommendations

5.1 Summary

The swift embrace of AI in educational assessment has ushered in a world of exciting opportunities, along with a host of complex ethical dilemmas. Automated assessment systems can offer great benefits in terms of efficiency, scalability and consistency, but the issues of transparency, fairness, accountability and inclusiveness need to be carefully considered.

This systematic literature review examined recent research on Explainable Artificial Intelligence (XAI) and its role in equity in automated educational assessment systems. The results of this study suggest that there are still concerns with algorithmic bias in particular when assessment models are trained on datasets that are not representative of the full spectrum of linguistic, cultural and socioeconomic populations. Such biases can lead to differential outcomes of assessment, and may disproportionately impact marginalized learners.

The review also shows the importance of Explainable Artificial Intelligence in terms of transparency by emphasizing the process of decision making for the assessment. Visualization techniques, feature attribution methods, and interpretable decision-support mechanisms can be employed to improve understanding of the automated assessment process. Increased transparency has been positively associated with increased trust, accountability and confidence of stakeholders;

But the evidence has been consistent that explainability doesn't alone ensure fairness. Transparency in an assessment decision does not prevent inequitable outcomes. The most promising approaches discussed in the literature are thus those that incorporate explainability and meaningful human oversight.

The most common model for ethical AI in education was Human-in-the-Loop. In such approaches the AI system is used as a decision support tool and the educator retains responsibility for the final decisions about assessment. These approaches allow the benefits of automation to be realised whilst maintaining the contextual judgement, empathy and professional expertise that continue to be central to educational assessment.

In conclusion, Explainable AI should be viewed as a core requirement for the responsible and fair use of AI for evaluation, not just as a technical add-on. In order for AI to be successfully applied in education it needs to be transparent, accountable, subject to human oversight and governed by institutions.

5.2 Recommendations for policy

Based on the findings reported in this review, some recommendations are made to education institutions, policy makers and technology developers.

Open-First Procurement Policies

Schools should be open about how they use AI to assess students. Careful consideration should be given to the deployment of systems that cannot deliver interpretable explanations for assessment decisions. Ask vendors to describe their process for generating automated scores and recommendations.

Requires Human Oversight

All high-stakes assessment activities should always be reviewed by humans. Automated systems might be used for preliminary scoring and feedback, but final academic decisions should be made by qualified instructors.

The following policy statement is being presented for adoption:

Artificial Intelligence tools may facilitate the assessment and feedback processes. Final decisions on academic standing, course grades, eligibility for admission or graduation will be made and approved by a human. In the event of inconsistencies or problems, human judgment will trump algorithm recommendations.

Auditing Algorithms

Regular audits should be carried out to check for possible bias in assessment systems. Audits should look at how AI models perform for different groups of students, different linguistic groups, and different education contexts.

Institutional AI Governance Frameworks

Educational institutions should establish multi-disciplinary governance committees to oversee the adoption and integration of AI. Committees must include representatives from academic departments, information technology units, administration and student government.

Student Visibility and Knowledge

Students should be made aware when they are using AI technologies for assessment or feedback. Transparency breeds trust, accountability and due process.

5.3 Pedagogical Suggestions

The findings of this review have several implications for teaching and learning practice.

Building Educator Capacity

Professional development programs should be developed to increase educators' knowledge of AI and explainability tools and algorithmic bias. Teachers need to be able to critically assess the AI's suggestions rather than take them at face value.

AI literacy of the students

Students need to know what they can and can't do with AI systems." Understanding how these automated assessment technologies work can help clarify misunderstandings and encourage the responsible use of AI-assisted learning environments.

Maintaining Learner Diversity

Assessment practices should value multiple forms of expression, communication and creativity. Educational technologies should facilitate and encourage cultural, linguistic and intellectual diversity, not hamper it.

6. Directions for Future Research

The review process has also identified several areas for future research.

First, more empirical work is needed in developing countries and in multilingual education contexts. Most research has been done in North America, Europe and East Asia and little is known about AI-assisted assessment in the Global South.

Second, future research should investigate the effectiveness of Explainable Artificial Intelligence in real classroom settings and not in controlled experimental settings. Longitudinal studies are needed to determine the long-term impact of AI-supported assessment on learning outcomes and educational equity. Third, more research is needed on the psychological effects of automated assessment. Little research exists on student motivation, trust, self-efficacy and academic identity.

Fourth, the effectiveness of AI systems in multi-lingual educational settings (e.g., settings where multiple languages and dialects are common) should be studied. Research of this type may be particularly beneficial

for schools in the Philippines and other multilingual countries.

Finally, future studies should investigate new explainability methods that can provide meaningful explanations while maintaining high predictive performance. One of the most salient issues in current AI research is the accuracy-interpretability trade-off.

7. Implications for Philippine Higher Education

This review has a great impact on the Higher Education Institutions in the Philippines. The increasing adoption of digital learning tools opens the door for AI-supported assessment systems to scale to large student populations and alleviate administrative burden.

But the Philippine educational setting is characterized by linguistic and cultural diversities, which present some challenges. Assessment systems trained primarily on international data may not identify local language practices, Philippine English variations, or culturally contextualized forms of communication. Hence, AI assisted assessment technologies need to be locally validated before their widespread use. Institutional policies and the adoption process should be driven by fairness, transparency and inclusion. One possible path forward for responsible innovation in education may be the intersection of explainable AI and human-in-the-loop frameworks. If well managed and transparently executed, technology could be an asset to educational institutions. If always humanly supervised, it could safeguard the principles of equity and academic integrity.

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